

Stochastic Volatility Models

(1. The Heston Model)

Stochastic Volatility Model Motivation

1. Motivation: Why Stochastic Volatility Models?

- Black-Scholes assumes one constant volatility number for all strikes and maturities. Real option markets disagree; implied volatility surfaces are not flat.

$$ds/s = rdt + \underbrace{\sigma(S,t)}_{\uparrow} dw$$



2. What Constant Volatility Fails to Explain

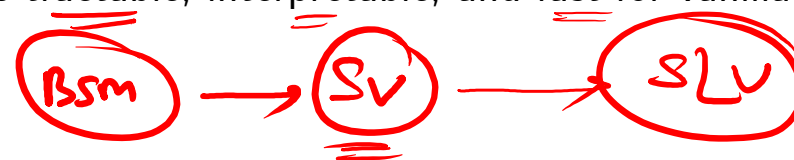
- Volatility smile**: Equity index options typically show higher implied volatility for lower strikes, reflecting downside crash risk.
- Volatility term structure**: Short-dated and long-dated options often trade at different implied vol levels
- Volatility clustering**: Large market moves are often followed by periods of elevated volatility.
- Leverage effect**: Equity spot declines are usually associated with rising volatility.

A stochastic volatility model treats volatility itself as a **dynamic risk factor**

Heston Model

Heston is widely used as a **benchmark stochastic volatility model** because it is tractable, interpretable, and fast for vanilla option pricing. Some features are

- Provides a clear link between **model parameters** and **smile behavior**
- Prices European options efficiently using **Fourier methods**
- Bridges the gap between **Black-Scholes** and more advanced models such as **stochastic local volatility** and **rough volatility**
- Makes key modelling dimensions explicit: volatility level, skew, curvature, term structure, and shock persistence



Under the risk-neutral measure $\mathbb{Q}$, the Heston model is commonly written as

$$\begin{aligned}
 ds &= rSdt + \underbrace{\sigma S}_{\sqrt{v_t}} dW^S \\
 \left\{ \begin{aligned} dS_t &= (r - q)S_t dt + \sqrt{v_t} S_t dW_t^S, \\ dv_t &= \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \\ d\langle W^S, W^v \rangle_t &= \rho dt \end{aligned} \right.
 \end{aligned}$$

GBM
CIR (square root)

($dW^S \cdot dW^v$) = ρdt

Here S_t is the spot price, v_t is the instantaneous variance, r is the continuously compounded risk-free rate, q is the continuous dividend yield or foreign rate, κ is the speed of variance mean reversion, θ is the long-run variance, ξ is the volatility of variance, and ρ is the spot-variance correlation.

Characteristic Function

Let $X_T = \log S_T$. Heston model is affine, so the characteristic function is given as

$$\phi(u; T) = \mathbb{E}^{\mathbb{Q}} [e^{iuX_T}] = \exp(C(u, T) + D(u, T)v_0 + iu \log S_0)$$

A commonly used form is

$$\begin{aligned} d(u) &= \sqrt{(\rho\xi iu - \kappa)^2 + \xi^2(iu + u^2)}, \\ g(u) &= \frac{\kappa - \rho\xi iu - d(u)}{\kappa - \rho\xi iu + d(u)}, \\ C(u, T) &= (r - q)iuT + \frac{\kappa\theta}{\xi^2} \left[(\kappa - \rho\xi iu - d)T - 2 \log \left(\frac{1 - ge^{-dT}}{1 - g} \right) \right], \\ D(u, T) &= \frac{\kappa - \rho\xi iu - d}{\xi^2} \left(\frac{1 - e^{-dT}}{1 - ge^{-dT}} \right). \end{aligned}$$

$$e^{A + Bx \text{ (initial values)}}$$

$$e^{A + B(x_0)}$$

$$T$$

FFT

European Option price by Fourier Inversion

For a European call with strike K and maturity T , the Heston price can be written as

$$C(S_0, K, T) = S_0 e^{-qT} P_1 - K e^{-rT} P_2$$

The probabilities P_1 and P_2 are not physical probabilities. They are Fourier inversion quantities

$$\begin{aligned} \rightarrow P_1 &= \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-iu \log K} \phi(u - i; T)}{iu \phi(-i; T)} \right] du, \\ \rightarrow P_2 &= \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-iu \log K} \phi(u; T)}{iu} \right] du. \end{aligned}$$

Put price follows from put-call parity

$$P(S_0, K, T) = C(S_0, K, T) - S_0 e^{-qT} + K e^{-rT}$$

Carr Madan FFT Price

The Carr-Madan method prices a whole grid of strikes by applying a Fourier transform to a damped call price.

Define log-strike $\underline{k} = \log K$.

The undiscounted call price as a function of k is not square integrable, so introduce a damping parameter $\alpha > 0$:

$$\underline{c_\alpha(k)} = \underline{e^{\alpha k} C(k)} \quad \alpha = 1.25 k$$

The Fourier transform of the damped call price is

$$\underline{\psi(v)} = \frac{e^{-rT} \phi(v - (\alpha + 1)i; T)}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v} \quad \leftarrow$$

The call price is recovered by inversion:

$$\underline{C(k)} = \frac{e^{-\alpha k}}{\pi} \int_0^\infty \text{Re} \left(e^{-ivk} \psi(v) \right) dv$$

Experiment



Implied volatility constructed out of these parameters

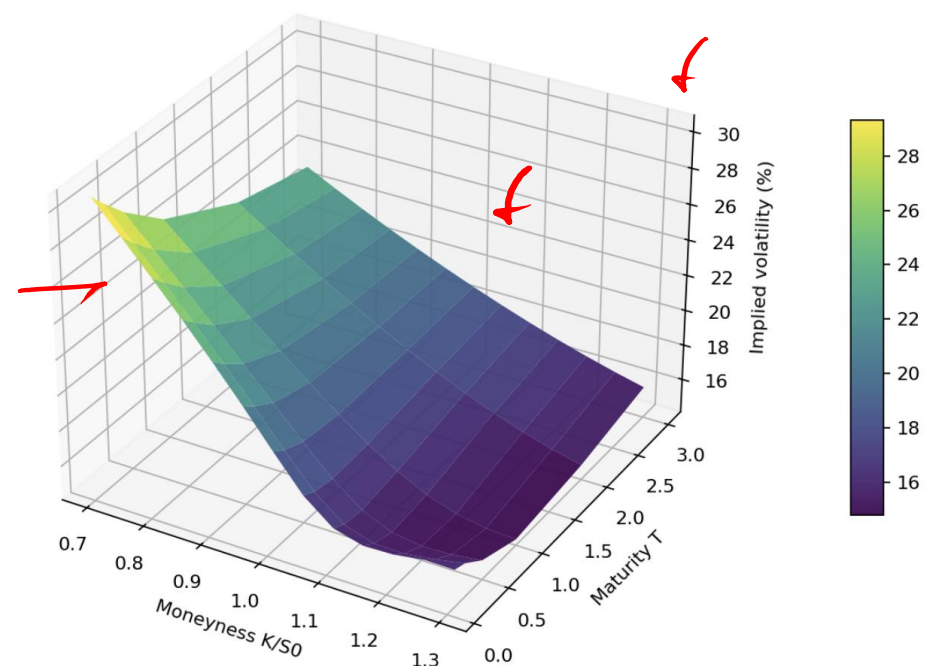
$$\underbrace{(v_0, \kappa, \theta, \xi, \rho)}_{\text{parameters}} = (\underbrace{0.0400}, \underbrace{1.50}, \underbrace{0.0400}, \underbrace{0.50}, \underbrace{-0.65}), \quad S_0 = \underbrace{100},$$

with $\underline{r = 4.5\%}$ and $\underline{q = 1.5\%}$

Heston price (FFT)

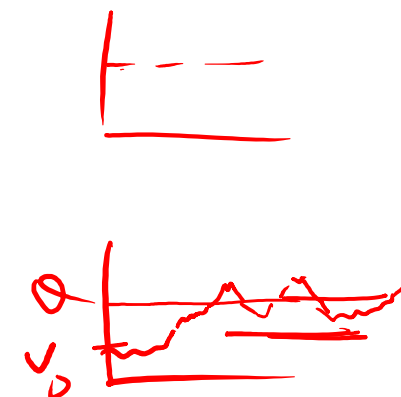
BSM-1
IV

$\sigma(S, t)$



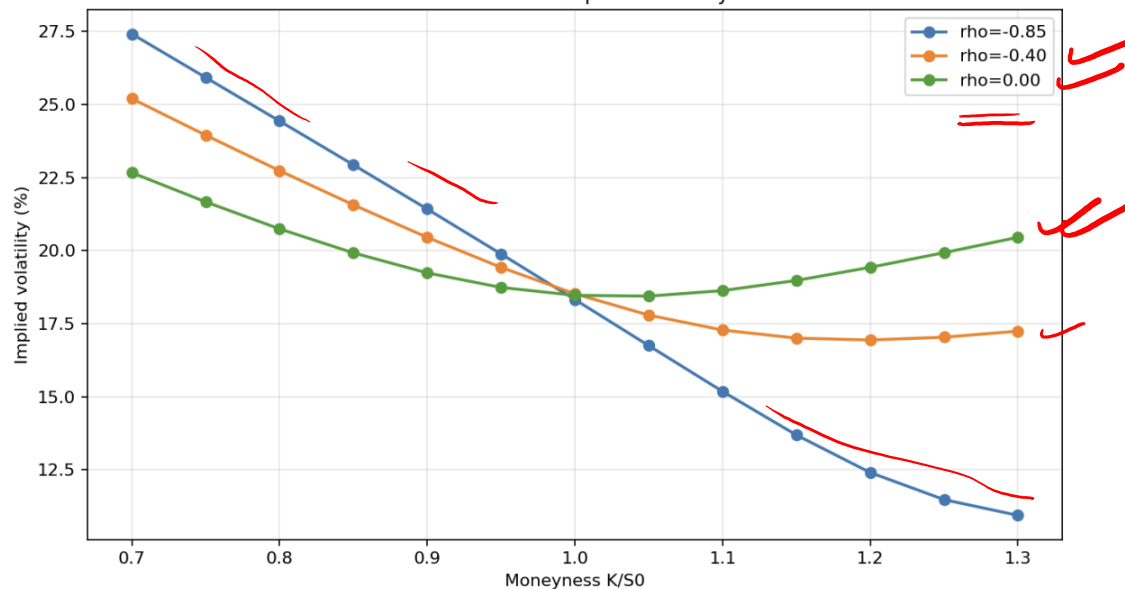
Heston parameters and smile dynamics

Parameter	Name	Practical interpretation
v_0	Initial variance	Controls the starting level of instantaneous variance. It has strongest effect on short-maturity ATM implied volatility.
θ	Long-run variance	Controls where variance tends to revert over long horizons. It has stronger effect on long-dated implied volatility.
κ	Mean-reversion speed	Controls how quickly variance moves from v_0 toward θ . Large κ makes variance shocks decay quickly. Small κ makes volatility regimes more persistent.
ξ	Vol-of-vol	Controls randomness of variance. Higher ξ increases smile curvature and tail thickness.
ρ	Spot-vol correlation	Controls smile slope. Negative ρ generates equity-like downside skew because spot declines are associated with variance increases.

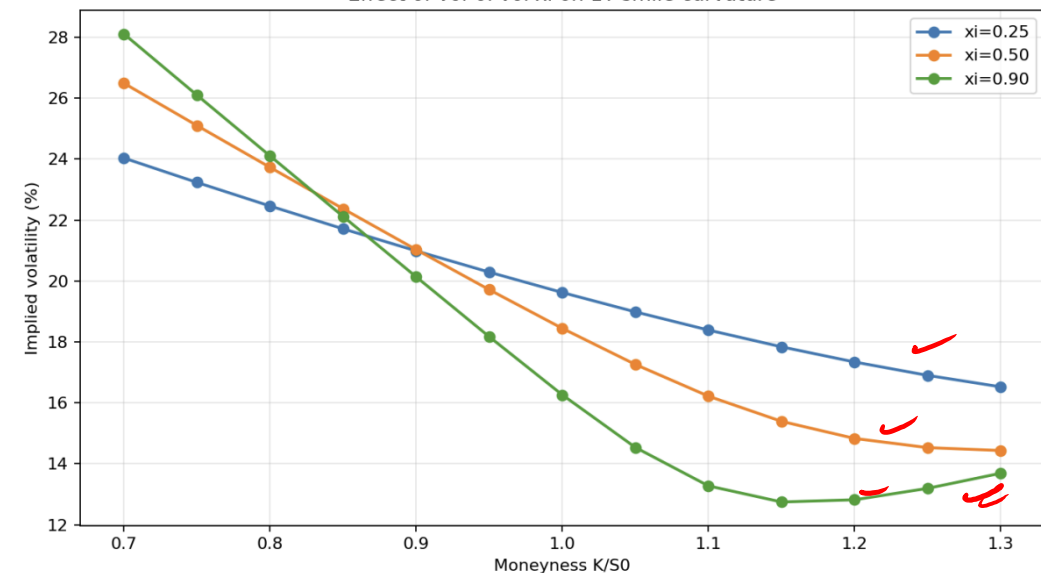


high persistence

Effect of parameters

Effect of ρ on 1Y implied-volatility skew

A more negative ρ steepens the left skew. Intuitively, when the spot falls, variance tends to rise; downside options become more valuable, raising low-strike implied volatility.

Effect of vol-of-vol ξ on 1Y smile curvature

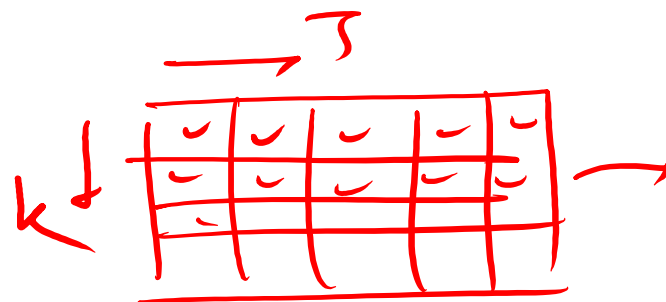
The parameter ξ controls the volatility of variance. Higher ξ makes future variance more uncertain. This increases convexity in option values and typically makes the implied-volatility smile more curved

Calibration



A production calibration begins with a liquid option chain: maturities, strikes, mid prices, bid-ask spreads, discount factors, dividends or forwards, and contract conventions.

1. Select liquid OTM calls and puts or convert all quotes to a consistent call price representation using put-call parity.
2. Convert market mid prices to implied volatilities
3. Choose parameter bounds and initial guesses.
4. Minimize either implied-vol errors
5. Validate the fit by maturity and strike



The objective function

$$\min_{\Theta} \sum_{i,j} w_{ij} \left(\sigma_{ij}^{\text{model}}(\Theta) - \sigma_{ij}^{\text{mkt}} \right)^2 \quad \text{where } \Theta = (v_0, \kappa, \theta, \xi, \rho)$$

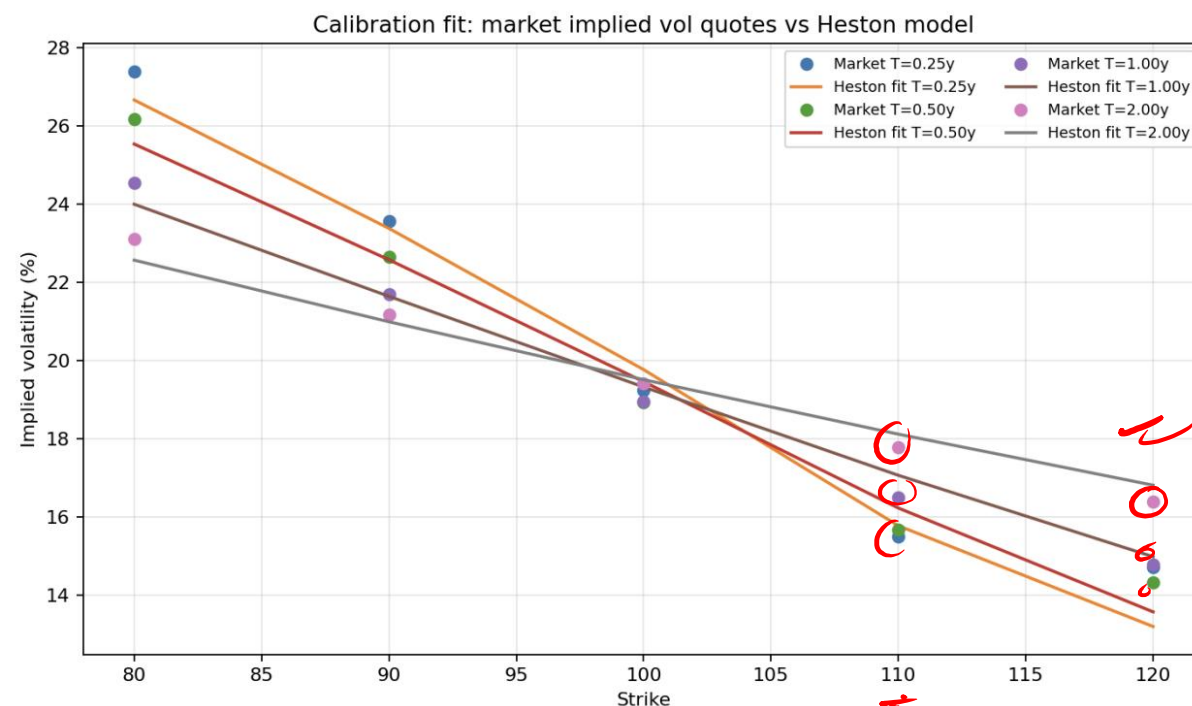
$\Theta = \{v_0, \kappa, \theta, \xi\}$

Calibration Results



Maturity	K=80	K=90	K=100	K=110	K=120
0.25Y	27.3870%	23.5631%	19.2419%	15.5023%	14.7288%
0.50Y	26.1702%	22.6554%	18.9482%	15.6801%	14.3441%
1.00Y	24.5389%	21.7050%	18.9681%	16.5096%	14.7837%
2.00Y	23.1096%	21.1837%	19.4104%	17.7925%	16.4035%

Parameter	Calibrated value
v_0	0.04146665
κ	2.16837938
θ	0.04103891
ξ	0.44288195
ρ	-0.82176845
Implied-volatility RMSE	0.5571 vol points



Heston Greeks

Because Heston has semi-closed-form Fourier prices, Greeks can be computed in several ways:

- **Finite differences.** Reprice after bumping an input. This is easiest to explain and is robust for a teaching implementation.
- **Analytical differentiation** of the characteristic function. Differentiate $C(u, T)$ and $D(u, T)$ with respect to parameters and integrate the derivatives. This is faster but algebraically more involved.
- **Algorithmic differentiation.** Useful in production libraries where many sensitivities must be calculated consistently.

For a call price $V(S_0, v_0, r, T; \Theta)$

$$\begin{aligned}
 \Delta &\approx \frac{V(S_0 + h_S) - V(S_0 - h_S)}{2h_S}, & \text{Center delta} \\
 \Gamma &\approx \frac{V(S_0 + h_S) - 2V(S_0) + V(S_0 - h_S)}{h_S^2}, \\
 \text{Vega}_{v_0} &\approx \frac{V(v_0 + h_v) - V(v_0 - h_v)}{2h_v}, & \frac{\partial V}{\partial v_0} \\
 \rho_r &\approx \frac{V(r + h_r) - V(r - h_r)}{2h_r}, \\
 \Theta_{1 \text{ day}} &\approx V(T - 1/365) - V(T).
 \end{aligned}$$

Quantity	Value
Call price	9.016722 ✓
Delta	0.677918 ✓
Gamma	0.017642
Vega to v_0	36.551572
Rate rho	58.775059
Theta per day	-0.013868

Pros and Cons

Advantages

- Smile generation. Unlike Black-Scholes, Heston naturally produces non-flat implied-volatility surfaces.
- Economic interpretability. Parameters map reasonably well to volatility level, skew, curvature, and term structure.
- Fast vanilla pricing. The characteristic function enables efficient Fourier and FFT pricing.
- Useful benchmark. Heston is a standard baseline for stochastic volatility discussions, model validation, and calibration exercises.
- Semi-analytical structure. The affine form supports analytical and numerical sensitivity calculations.

Limitations

- Limited short-dated skew flexibility. The model may struggle with very steep short-dated smiles observed after market shocks.
- Calibration instability. Different parameter sets can produce similar vanilla prices. This creates identifiability issues.
- No jumps in the basic version. Large discontinuous moves are not captured unless jumps are added.
- Vanilla fit is not enough. A model calibrated to vanillas may still misprice exotics if the dynamics are wrong.
- Numerical care required. Branch cuts, FFT grid choices, interpolation, and implied-vol inversion can all introduce errors.